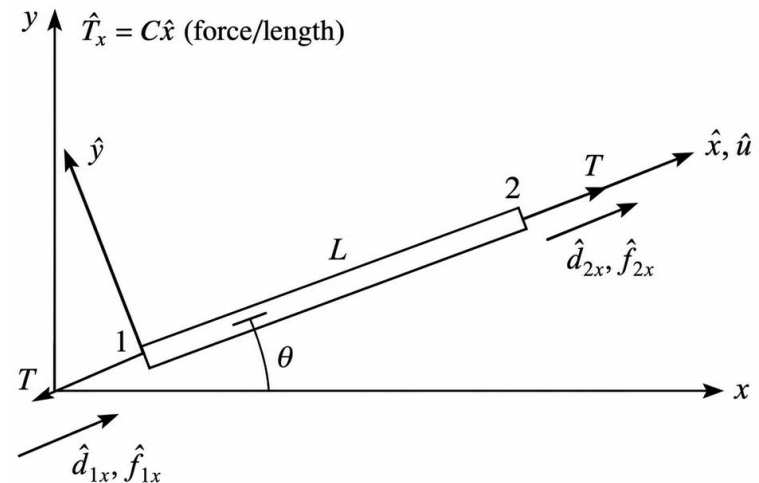


Bar Element

- Derivation of stiffness matrix for a bar element in local coordinate system

- Linear elastic
- Constant cross section
- Tensile force T along the axis at nodes 1 and 2
- Local coordinate system $(\hat{x}, \hat{y})$
- Global coordinate system (x, y)
- Cross section A , modulus E , length L



Bar Element

- Hooke's law $\sigma_x = E \varepsilon_x$

$$\varepsilon_x = \frac{d\hat{u}}{d\hat{x}}$$

- From force equilibrium

$$A\sigma_x = T = \text{constant} \quad T = AE \frac{d\hat{u}}{d\hat{x}} = \text{Constant}$$

- Governing equation

$$\frac{d}{d\hat{x}} \left(AE \frac{d\hat{u}}{d\hat{x}} \right) = 0$$

$\hat{u}$ = axial displacement function along the element in $\hat{x}$ direction



Bar Element

- Assumptions in deriving the bar element stiffness matrix

- The bar cannot sustain shear force or bending moment

$$\hat{f}_{1y} = 0, \quad \hat{f}_{2y} = 0, \quad \hat{m}_1 = 0 \quad \text{and} \quad \hat{m}_2 = 0,$$

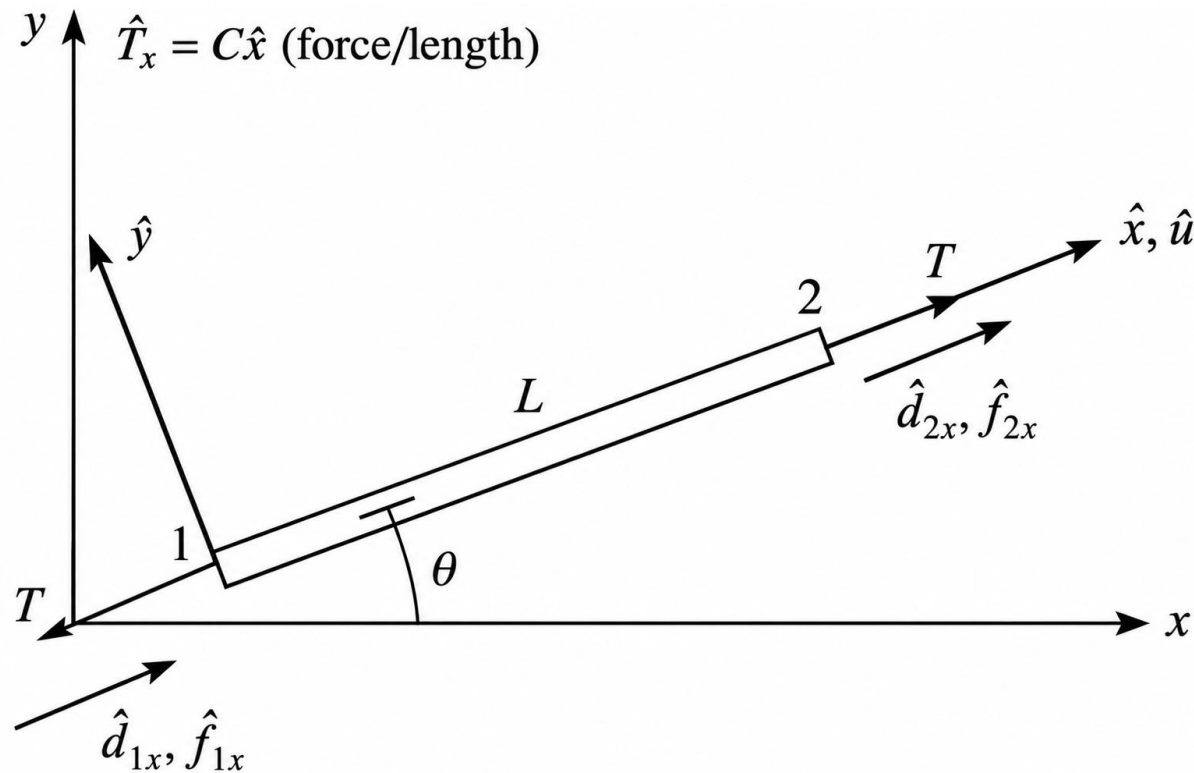
- Any effect of transverse displacement is ignored
- Hook's law applies

$$\sigma_x = E \varepsilon_x$$

- No intermediate applied load (only nodal forces)

Bar Element

- Step 1 : Select the element type assign node and element numbers



Bar Element

- **Step 2: Select a displacement function**
 - Assume a linear displacement variation along $\hat{x}$ axis
 - Linear function with specified end points has a unique path
 - Endpoints are nodal displacements

$$\hat{u} = a_1 + a_2 \hat{x} \qquad \hat{d}_{1x} \text{ and } \hat{d}_{2x}$$

- Two nodes in bar element
- Using the same procedure as in the spring derivation

$$\hat{u} = \left(\frac{\hat{d}_{2x} - \hat{d}_{1x}}{L} \right) \hat{x} + \hat{d}_{1x} \qquad \hat{u} = \begin{bmatrix} N_1 & N_2 \end{bmatrix} \begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{2x} \end{Bmatrix}$$

Bar Element

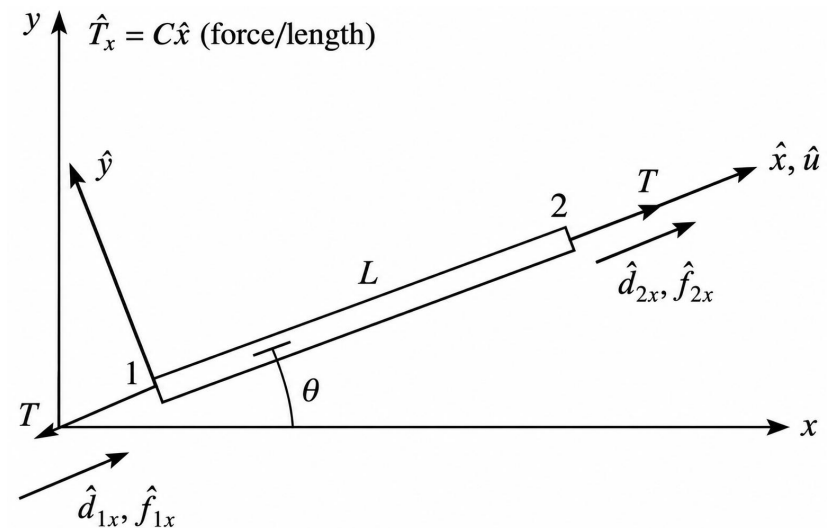
- Step 3: Define strain/displacement and stress/strain relationships

- Strain/displacement relationship

$$\varepsilon_x = \frac{d\hat{u}}{d\hat{x}} = \frac{\hat{d}_{2x} - \hat{d}_{1x}}{L}$$

- Stress/strain relationship

$$\sigma_x = E\varepsilon_x$$



Bar Element

- Step 4: Derive the element stiffness matrix and equations

$$T = A\sigma_x \quad \Rightarrow \quad T = AE\varepsilon_x \quad \Rightarrow \quad T = AE\left(\frac{\hat{d}_{2x} - \hat{d}_{1x}}{L}\right)$$

– From the sign convention

$$\hat{f}_{1x} = -T = \frac{AE}{L}(\hat{d}_{1x} - \hat{d}_{2x})$$

$$\hat{f}_{2x} = T = \frac{AE}{L}(\hat{d}_{2x} - \hat{d}_{1x})$$



Bar Element

- In the matrix form

$$\begin{Bmatrix} \hat{f}_{1x} \\ \hat{f}_{2x} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{2x} \end{Bmatrix}$$

$$\{\hat{f}\} = [\hat{k}]\{\hat{d}\}$$

$$[\hat{k}] = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Stiffness matrix of a bar element
in local coordinate system



Bar Element

- Step 5: Assemble element equations to obtain global or total equations

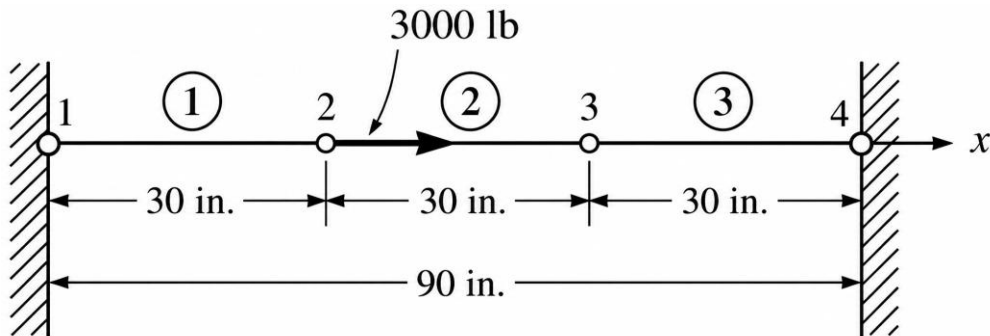
$$[K] = \sum_{e=1}^N [k]^{(e)} \quad \text{and} \quad \{F\} = \sum_{e=1}^N \{f\}^{(e)}$$

- All local stiffness $[k]$ Matrices must be transformed to global $[K]$
- Step 6: Solve for nodal displacements
 - Impose boundary conditions and solve the system of equations $\{F\} = [K]\{d\}$
- Step 7: Solve for elemental forces
 - Determine stress and strain in each element by back substitution of displacements



Bar Element

- For a three bar assemblage



- $E=30 \times 10^6$ psi and $A=1$ in² for elements 1 and 2
- $E=15 \times 10^6$ psi and $A=2$ in² for element 3
- Nodes 1 and 4 are fixed

Determine:

- Global stiffness matrix
- The displacements of nodes 2 and 3
- Reactions at nodes 1 and 4

Bar Element

(a)

$$\hat{k} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

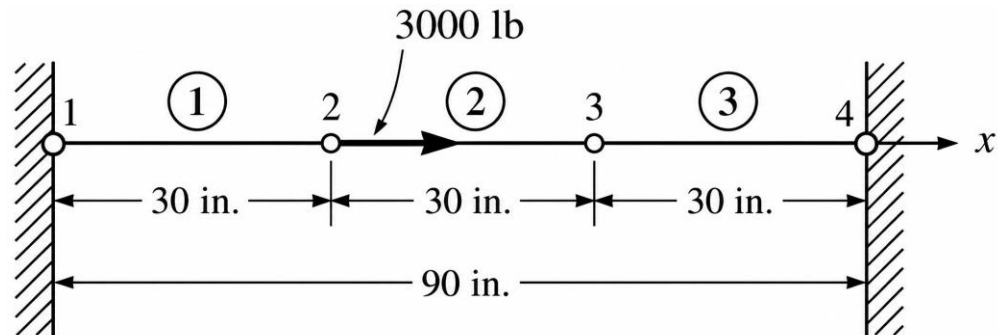
$$[k]^{(1)} = [k]^{(2)} = \frac{1 \times 30 \times 10^6}{30} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 10^6 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{ lb/in}$$

$\begin{matrix} 1 & 2^{(1)} \\ 2 & 3^{(2)} \end{matrix}$

$$[k]^{(3)} = \frac{2 \times 15 \times 10^6}{30} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 10^6 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{ lb/in}$$

$\begin{matrix} 3 & 4 \\ 3 & 4 \end{matrix}$

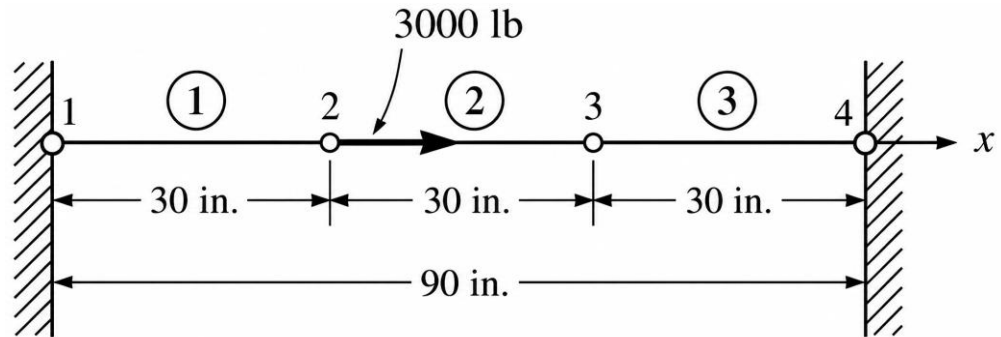
$$[K] = 10^6 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \text{ lb/in}$$



Bar Element

(b)

$$\begin{Bmatrix} F_{1x} \\ F_{2x} \\ F_{3x} \\ F_{4x} \end{Bmatrix} = [K] \begin{Bmatrix} d_{1x} \\ d_{2x} \\ d_{3x} \\ d_{4x} \end{Bmatrix}$$



$d_{1x} = d_{4x} = 0$: Boundary conditions

$$\therefore \begin{Bmatrix} 3000 \\ 0 \end{Bmatrix} = 10^6 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{Bmatrix} d_{2x} \\ d_{3x} \end{Bmatrix}$$

$$d_{2x} = 0.002" \quad d_{3x} = 0.001"$$

$$(c) \quad F_{1x} = 10^6(d_{1x} - d_{2x}) = 10^6(0 - 0.002) = -2000lb$$

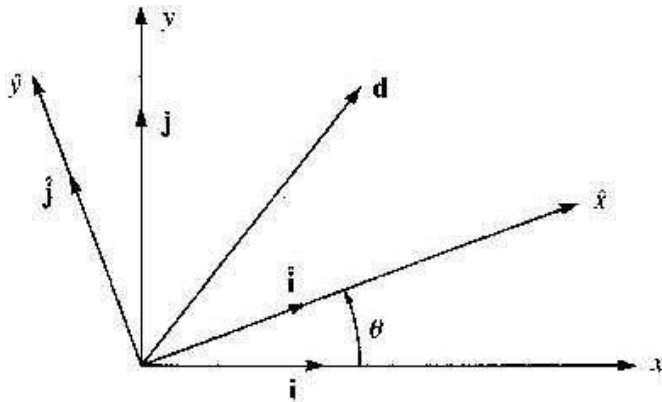
$$F_{2x} = 10^6(-d_{1x} + 2d_{2x} - d_{4x}) = 10^6(0 + 2 \times 0.002 - 0.001) = -3000lb$$

$$F_{3x} = 10^6(-d_{2x} + 2d_{3x} - d_{4x}) = 10^6(-0.002 + 2 \times 0.001) = 0lb$$

$$F_{4x} = 10^6(-d_{3x} + d_{4x}) = 10^6(-0.001 + 0) = -1000lb$$

Bar Element

- Transformation of vectors in 2D for conversion from local to global coordinate system



d = nodal displacement of an element
 θ = positive CCW from x to $\hat{x}$

- Develop a transformation matrix to relate components of d from one coordinate system to another

Coordinate Transformation

$$d = d_x i + d_y j = \hat{d}_x \hat{i} + \hat{d}_y \hat{j}$$

$$a + b = i$$

$$a = (\cos \theta) \hat{i}$$

$$b = (\sin \theta)(-\hat{j})$$

$$\hat{i} = (\cos \theta) \hat{i} - (\sin \theta) \hat{j}$$

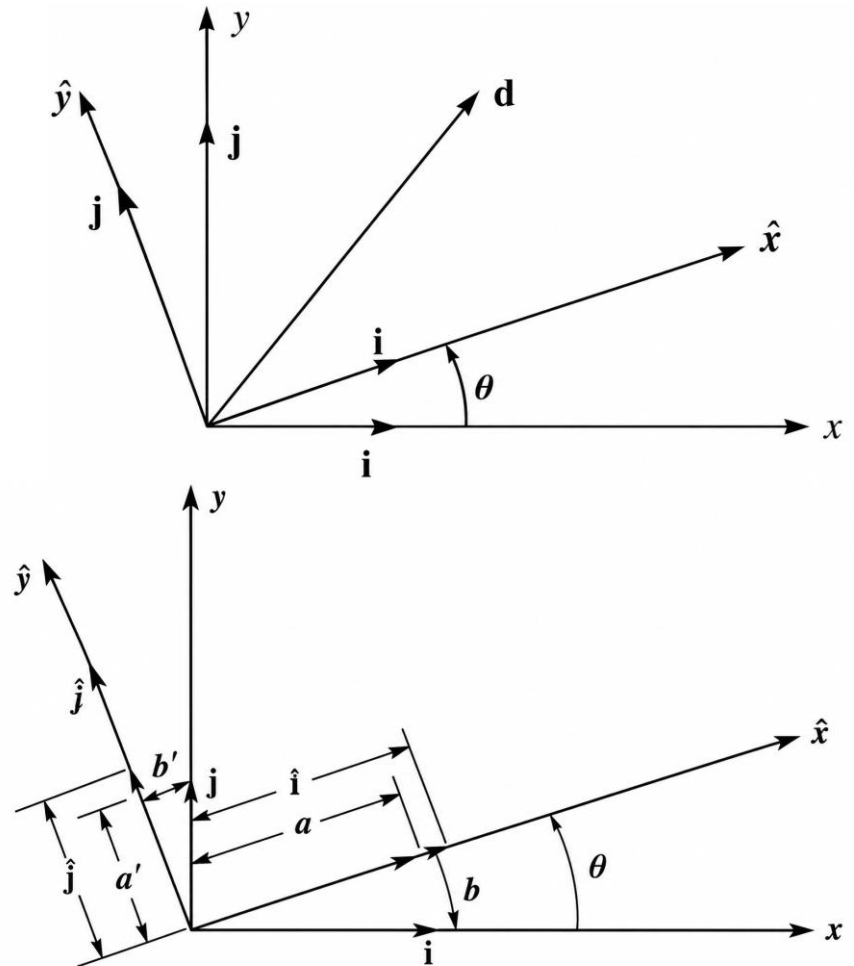
Similarly

$$a' + b' = j$$

$$a' = (\cos \theta) \hat{j}$$

$$b' = (\sin \theta)(\hat{i})$$

$$\hat{j} = (\sin \theta) \hat{i} + (\cos \theta) \hat{j}$$



Coordinate Transformation

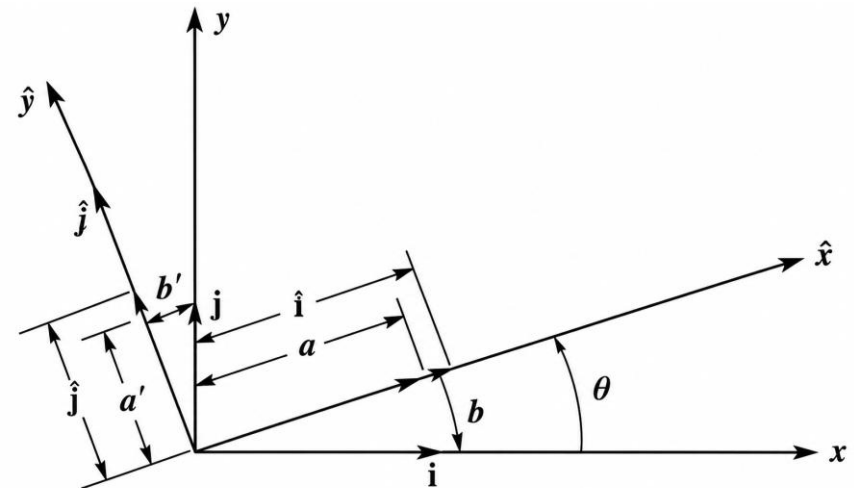
$$d_x(\cos \theta \hat{i} - \sin \theta \hat{j}) + d_y(\sin \theta \hat{i} + \cos \theta \hat{j}) = \hat{d}_x \hat{i} + \hat{d}_y \hat{j}$$

$$\left. \begin{aligned} \text{or } d_x \cos \theta + d_y \sin \theta &= \hat{d}_x \\ -d_x \sin \theta + d_y \cos \theta &= \hat{d}_y \end{aligned} \right\}$$

combine like
coefficients

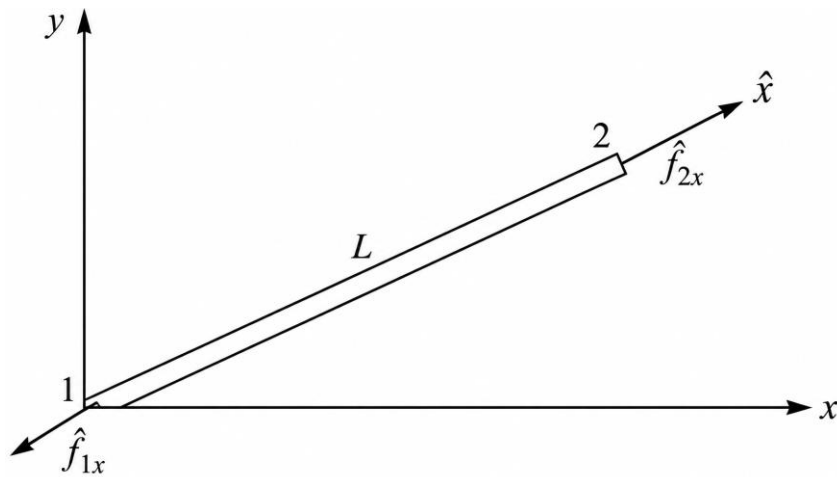
$$\begin{Bmatrix} \hat{d}_x \\ \hat{d}_y \end{Bmatrix} = \begin{bmatrix} C & S \\ -S & C \end{bmatrix} \begin{Bmatrix} d_x \\ d_y \end{Bmatrix}$$

Transformation matrix



Bar Element

- Conversion from local to global coordinate system
 - In the local coordinate system for a bar element



$$\begin{Bmatrix} \hat{f}_{1x} \\ \hat{f}_{2x} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{2x} \end{Bmatrix} \Rightarrow \{\hat{f}\} = [\hat{k}]\{\hat{d}\}$$

Bar element arbitrarily oriented w.r.t. x-y.

Means we want to find a matrix $[k]$ such that

$$\begin{Bmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \end{Bmatrix} = [k] \begin{Bmatrix} d_{1x} \\ d_{1y} \\ d_{2x} \\ d_{2y} \end{Bmatrix} \quad \{f\} = [k]\{d\}$$

- In local coordinate system there are only two force and displacement components
- In global coordinate system there are four force and displacement components

Bar Element

$$\left. \begin{aligned} \hat{d}_{1x} &= d_{1x} \cos \theta + d_{1y} \sin \theta \\ \hat{d}_{2x} &= d_{2x} \cos \theta + d_{2y} \sin \theta \end{aligned} \right\} \Rightarrow \begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{2x} \end{Bmatrix} = \begin{bmatrix} C & S & 0 & 0 \\ 0 & 0 & C & S \end{bmatrix} \begin{Bmatrix} d_{1x} \\ d_{1y} \\ d_{2x} \\ d_{2y} \end{Bmatrix}$$

$$\{\hat{d}\} = [T]^* \{d\}$$

Similarly,

$$\begin{Bmatrix} \hat{f}_{1x} \\ \hat{f}_{2x} \end{Bmatrix} = [T]^* \begin{Bmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \end{Bmatrix} \Rightarrow \{\hat{f}\} = [k]\{\hat{d}\} = [\hat{k}][T]^* \{d\}$$

$$[T]^* \{f\} = [\hat{k}][T]^* \{d\}$$

Bar Element

- Now $[T]^*$ should be inverted to solve this equation
- But $[T]^*$ is not square matrix and needs enhancement

$$\begin{Bmatrix} \hat{d}_x \\ \hat{d}_y \end{Bmatrix} = \begin{bmatrix} C & S \\ -S & C \end{bmatrix} \begin{Bmatrix} d_x \\ d_y \end{Bmatrix}$$

$$\begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{1y} \\ \hat{d}_{2x} \\ \hat{d}_{2y} \end{Bmatrix} = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \begin{Bmatrix} d_{1x} \\ d_{1y} \\ d_{2x} \\ d_{2y} \end{Bmatrix}$$

$$\{\hat{d}\} = [T]\{d\}$$

Similarly,

$$\{\hat{f}\} = [T]\{f\}$$



Bar Element

$\begin{bmatrix} \hat{k} \end{bmatrix}$ must also be expanded to 4×4 matrix

$$\begin{Bmatrix} \hat{f}_{1x} \\ \hat{f}_{1y} \\ \hat{f}_{2x} \\ \hat{f}_{2y} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{1y} \\ \hat{d}_{2x} \\ \hat{d}_{2y} \end{Bmatrix}$$

$$\{\hat{f}\} = [\hat{k}]\{\hat{d}\}$$

$$[T]\{f\} = [\hat{k}][T]\{d\} \quad \Rightarrow \quad \{f\} = [T]^{-1}[\hat{k}][T]\{d\}$$

However, $[T]^{-1} = [T]^T$

$$\{f\} = [T]^T [\hat{k}][T]\{d\}$$

$$[k]\{d\} = [T]^T [\hat{k}][T]\{d\}$$

$$[k] = [T]^T [\hat{k}][T]$$



Bar Element

$$[k] = [T]^T [\hat{k}] [T]$$

$$[k] = \frac{AE}{L} \begin{bmatrix} C^2 & CS & -C^2 & -CS \\ & S^2 & -CS & -S^2 \\ & & C^2 & CS \\ \text{symm} & & & S^2 \end{bmatrix}$$

$$[k] = \frac{AE}{L} \begin{bmatrix} \lambda & -\lambda \\ -\lambda & \lambda \end{bmatrix}$$



Bar Element

- The displacement function $\hat{u} = a_1 + a_2\hat{x}$ was assumed piecewise continuous element by element
- The stiffness matrix for each element can be summed by using direct stiff method

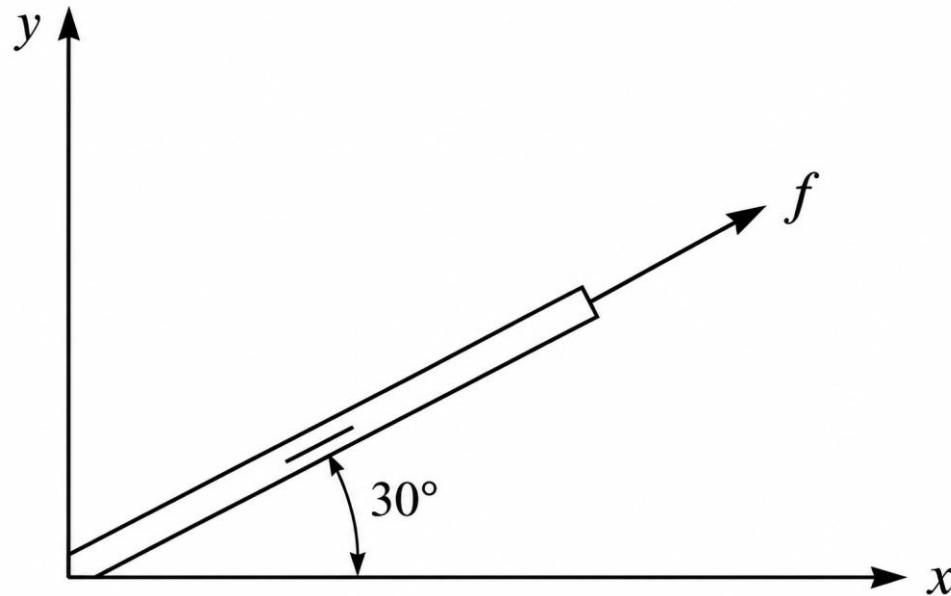
$$\sum_{e=1}^N [k]^{(e)} = [K] \quad \leftarrow \begin{array}{l} \text{Total stiffness matrix} \\ N = \text{number of elements} \end{array}$$

– Similarly each global nodal force matrix:

$$\sum_{e=1}^N \{f\}^{(e)} = \{F\}$$
$$\therefore \{F\} = [K]\{d\} \quad \text{for the whole structure}$$

Bar Element

- For the bar element shown
 - Evaluate global stiffness matrix w.r.t. x - y coordinate system
 - $A=2 \text{ in}^2$, $L=60''$ and $E=30 \times 10^6 \text{ psi}$



Bar Element

- Global stiffness matrix $[K]$

$$[k] = \frac{AE}{L} \begin{bmatrix} C^2 & CS & -C^2 & -CS \\ & S^2 & -CS & -S^2 \\ & & C^2 & CS \\ \text{symm} & & & S^2 \end{bmatrix}$$

$$\theta = 30^\circ, \quad C = \cos 30^\circ = \frac{\sqrt{3}}{2}, \quad S = \sin 30^\circ = \frac{1}{2}$$

$$[k] = \frac{2 \times 30 \times 10^6}{60} \begin{bmatrix} 0.75 & 0.433 & -0.75 & -0.433 \\ & 0.25 & -0.433 & -0.25 \\ & & 0.75 & 0.433 \\ & & & 0.25 \end{bmatrix} lb / in$$



Bar Element

- Stress in a bar in the X - Y plane

$$\begin{Bmatrix} \hat{f}_{1x} \\ \hat{f}_{2x} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{2x} \end{Bmatrix} \quad \text{Local forces related to local displacement}$$

Axial tensile stress $\sigma = \frac{\hat{f}_{2x}}{A}$,

$$\hat{f}_{2x} = \frac{AE}{L} \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{2x} \end{Bmatrix}$$

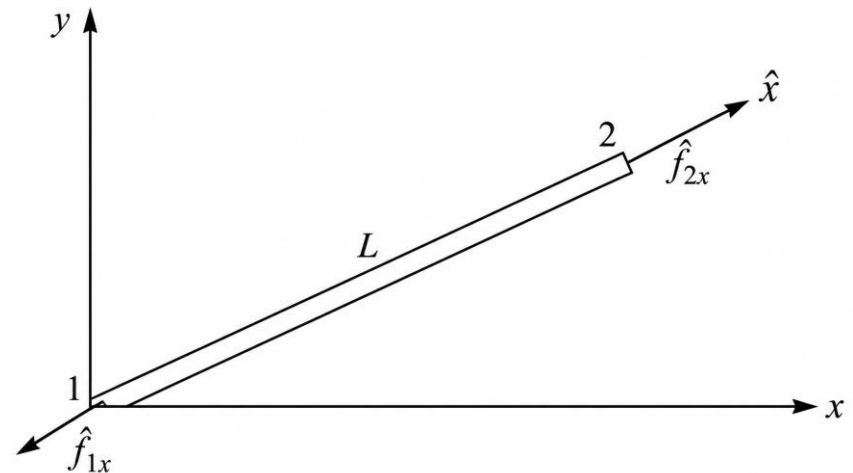
$$\sigma = \frac{E}{L} \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{Bmatrix} \hat{d}_{1x} \\ \hat{d}_{2x} \end{Bmatrix} = \frac{E}{L} \begin{bmatrix} -1 & 1 \end{bmatrix} \{\hat{d}\}$$

$$\sigma = \frac{E}{L} \begin{bmatrix} -1 & 1 \end{bmatrix} [T]^* \{d\}$$

or $\sigma = [C'] \{d\}$

Where, $[C'] = \frac{E}{L} \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} C & S & 0 & 0 \\ 0 & 0 & C & S \end{bmatrix}$

$$[C'] = \frac{E}{L} \begin{bmatrix} -C & -S & C & S \end{bmatrix}$$



Bar Element

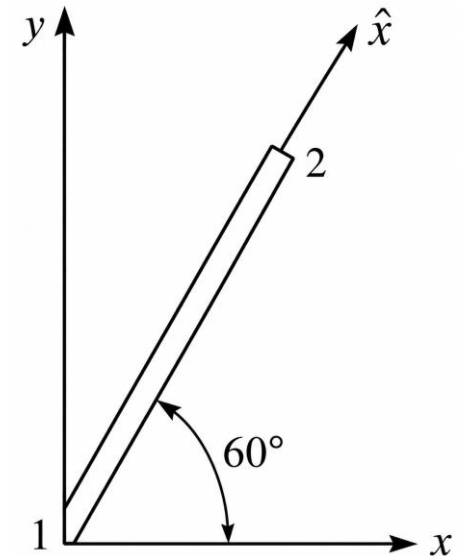
- Determine the axial stress

$$A = 4 \times 10^{-4} \text{ m}^2, \quad E = 210 \text{ GPa}, \quad L = 2 \text{ m}$$

- Global displacement is give as

$$d_{1x} = 0.25 \text{ mm}, \quad d_{2x} = 0.5 \text{ mm}$$

$$d_{1y} = 0.0 \text{ mm}, \quad d_{2y} = 0.75 \text{ mm}$$



$$\theta = 60^\circ$$

$$C = \cos 60^\circ = \frac{1}{2}$$

$$S = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

Bar Element

$$[C'] = \frac{E}{L} \begin{bmatrix} -C & -S & C & S \end{bmatrix}$$

$$[C'] = \frac{210 \times 10^6}{2} \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \frac{kN / m^2}{m}$$

$$\{d\} = \begin{Bmatrix} d_{1x} \\ d_{1y} \\ d_{2x} \\ d_{2y} \end{Bmatrix} = \begin{Bmatrix} 0.25 \\ 0 \\ 0.5 \\ 0.75 \end{Bmatrix} \times 10^{-3} m$$

$$\{\sigma_x\} = [C']\{d\}$$

$$\{\sigma_x\} = \frac{210 \times 10^6}{2} \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{Bmatrix} 0.25 \\ 0 \\ 0.5 \\ 0.75 \end{Bmatrix} \times 10^{-3}$$

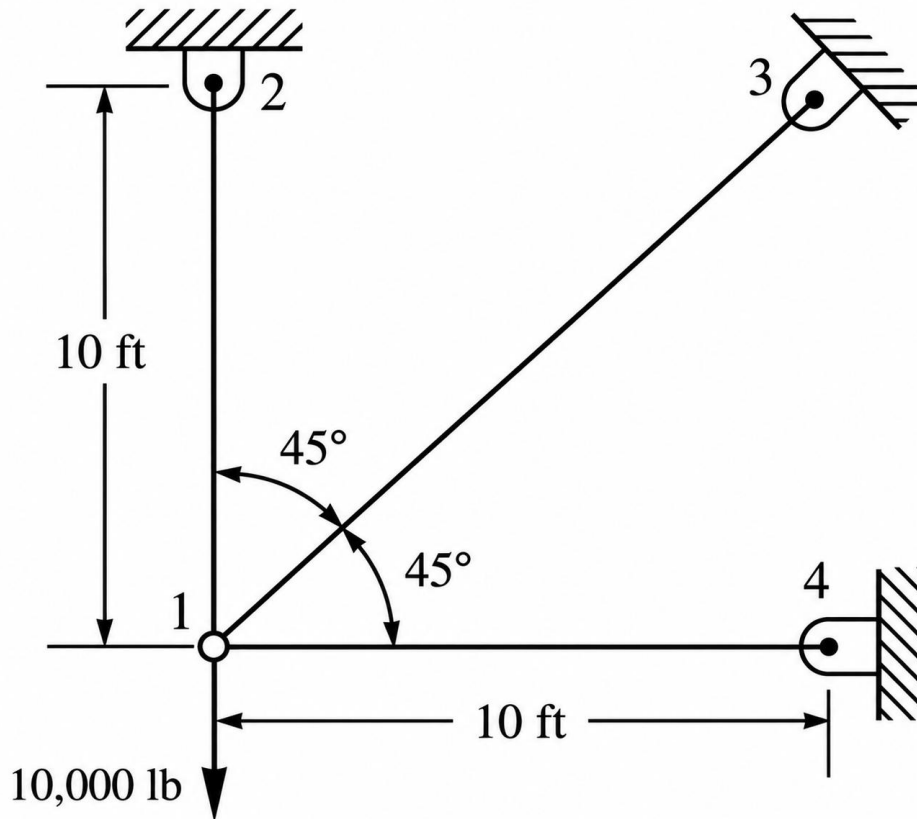
$$\{\sigma_x\} = 81.32 \times 10^3 kN / m^2 = 81.32 MPa$$

Bar Element

- **Solution of a plane truss:**
 - Using equations for bars in direct stiffness method
 - Plane truss: structure composed of bar elements that all lie in a common plane
 - Connected by frictionless pins
 - All the loads are also acting on the common plane and are applied on nodes or joints.



Bar Element



Determine x and y displacement of node 1 and stress in each element. $E=30 \times 10^6 \text{ psi}$, $A=2 \text{ in}^2$

Bar Element

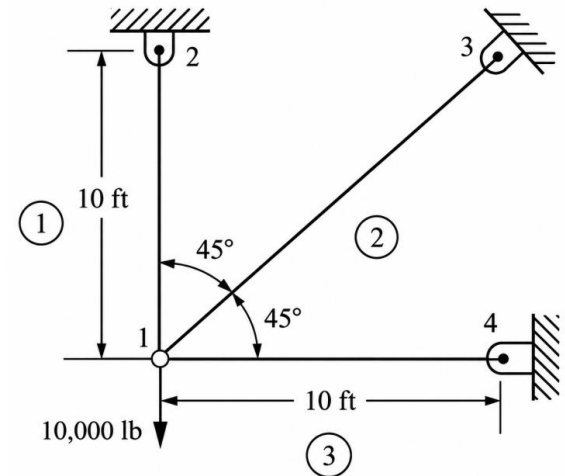
- Determine global stiffness matrix
- Angles are measured

between X and $\hat{X}$

Element 1: $\theta^{(1)} = 90^\circ$

Element 2: $\theta^{(2)} = 45^\circ$

Element 3: $\theta^{(3)} = 0^\circ$



- Total 8 nodal components of displacements (8 DoFs)

Element	θ°	C	S	C^2	S^2	CS
1	90°	0	1	0	1	0
2	45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
3	0°	1	0	1	0	0

Bar Element

$$[k]^{(1)} = \frac{AE}{L} \begin{bmatrix} C^2 & CS & -C^2 & -CS \\ & S^2 & -CS & -S^2 \\ & & C^2 & CS \\ \text{symm} & & & S^2 \end{bmatrix} = \frac{30 \times 10^6 \times 2}{120} \begin{bmatrix} d_{1x} & d_{1y} & d_{2x} & d_{2y} \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{matrix} d_{1x} \\ d_{1y} \\ d_{2x} \\ d_{2y} \end{matrix} \quad \text{Element 1}$$

$$[k]^{(2)} = \frac{30 \times 10^6 \times 2}{120 \times \sqrt{2}} \begin{bmatrix} d_{1x} & d_{1y} & d_{3x} & d_{3y} \\ 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix} \begin{matrix} d_{1x} \\ d_{1y} \\ d_{3x} \\ d_{3y} \end{matrix} \quad \text{Element 2}$$

$$[k]^{(3)} = \frac{30 \times 10^6 \times 2}{120} \begin{bmatrix} d_{1x} & d_{1y} & d_{4x} & d_{4y} \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} d_{1x} \\ d_{1y} \\ d_{4x} \\ d_{4y} \end{matrix} \quad \text{Element 3}$$

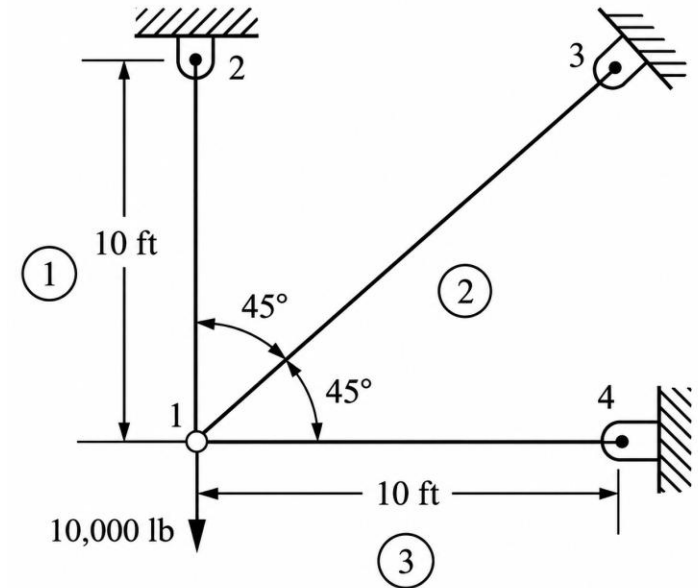
Bar Element

$$[K] = 500,000 \begin{bmatrix} d_{1x} & d_{1y} & d_{2x} & d_{2y} & d_{3x} & d_{3y} & d_{4x} & d_{4y} \\ 1.354 & 0.345 & 0 & 0 & -0.354 & -0.354 & -1 & 0 \\ & 1.354 & 0 & -1 & -0.354 & -0.354 & 0 & 0 \\ & & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & 1 & 0 & 0 & 0 & 0 \\ & & & & 0.354 & 0.354 & 0 & 0 \\ & & & & & 0.354 & 0 & 0 \\ & & & & & & 1 & 0 \\ & & & & & & & 0 \end{bmatrix}$$

Symm

$$\begin{Bmatrix} 0 \\ -10,000 \\ F_{2x} \\ F_{2y} \\ F_{3x} \\ F_{3y} \\ F_{4x} \\ F_{4y} \end{Bmatrix} = [K] \begin{Bmatrix} d_{1x} \\ d_{1y} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \Rightarrow \begin{Bmatrix} 0 \\ -10,000 \end{Bmatrix} = 50,000 \begin{bmatrix} 1.354 & 0.354 \\ 0.354 & 1.354 \end{bmatrix} \begin{Bmatrix} d_{1x} \\ d_{1y} \end{Bmatrix}$$

$$d_{1x} = 0.414 \times 10^{-2} \text{ in}; \quad d_{1y} = -1.59 \times 10^{-2} \text{ in}$$



Bar Element

$$\{\sigma\} = [C']\{d\};$$

$$[C'] = \frac{E}{L} \begin{bmatrix} -C & -S & C & S \end{bmatrix}$$

$$\sigma^{(1)} = \frac{30 \times 10^6}{120} \begin{bmatrix} 0 & -1 & 0 & 1 \end{bmatrix} \begin{Bmatrix} d_{1x} = 0.414 \times 10^{-2} \\ d_{1y} = -1.59 \times 10^{-2} \\ d_{2x} = 0 \\ d_{2y} = 0 \end{Bmatrix} = 3965 \text{ psi}$$

$$\sigma^{(2)} = \frac{30 \times 10^6}{120\sqrt{2}} \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{Bmatrix} d_{1x} = 0.414 \times 10^{-2} \\ d_{1y} = -1.59 \times 10^{-2} \\ d_{3x} = 0 \\ d_{3y} = 0 \end{Bmatrix} = 1471 \text{ psi}$$

$$\sigma^{(3)} = \frac{30 \times 10^6}{120} \begin{bmatrix} -1 & 0 & 1 & 0 \end{bmatrix} \begin{Bmatrix} d_{1x} = 0.414 \times 10^{-2} \\ d_{1y} = -1.59 \times 10^{-2} \\ d_{4x} = 0 \\ d_{4y} = 0 \end{Bmatrix} = -1035 \text{ psi}$$

Verify That $\sum F_x = 0 \Rightarrow 1471 \times 2 \times \frac{\sqrt{2}}{2} - 1035 \times 2 = 0$

$$\sum F_y = 0 \Rightarrow 3965 \times 2 + 1471 \times 2 \times \frac{\sqrt{2}}{2} - 10,000 = 0$$

